Design the Twitch API

SystemsExpert design walkthrough

Clarifying questions

- Which parts of the page are in scope: video player, chat, viewer count, follow/subscribe buttons?
- Logged-in and anonymous viewers?
- Are there chat moderation features (bans, slow mode, emote-only)?
- Does the API serve the video itself, or only metadata plus a playback URL?

Functional requirements

- Get stream info for a channel: title, game, live/offline, viewer count, playback URL
- Real-time chat: send and receive messages, with moderation
- Follow and unfollow a channel; subscribe (paid tiers)
- Notify followers when a channel goes live

Non-functional requirements

- Chat fan-out to 100k+ viewers in one channel, delivered in under a second
- Viewer counts can be approximate and a few seconds stale
- Subscriptions and payments must be strongly consistent and idempotent

Entities

```
Channel   { id, ownerUserId, title, gameId, isLive, streamStartedAt }
User      { id, username, ... }
Follow    { userId, channelId, createdAt }
Subscription { id, userId, channelId, tier, status, renewsAt }
ChatMessage  { id, channelId, userId, text, createdAt }
Ban       { channelId, userId, expiresAt? }
```

API

```
GET    /channels/{channelId}                -> Channel + viewerCount + playbackUrl (signed, short-lived)
GET    /channels/{channelId}/viewers/count
POST   /channels/{channelId}/follow         (idempotent)
DELETE /channels/{channelId}/follow
POST   /channels/{channelId}/subscriptions { tier, paymentToken, idempotencyKey }
DELETE /subscriptions/{subscriptionId}

# Chat: long-lived connection
WS     /channels/{channelId}/chat
  client -> { type: "send", text, clientMessageId }
  server -> { type: "message", ChatMessage } | { type: "moderation", action, userId }
POST   /channels/{channelId}/bans { userId, durationSec? }   (moderators only)
POST   /channels/{channelId}/chat-settings { slowModeSec, emoteOnly, followersOnly }
```

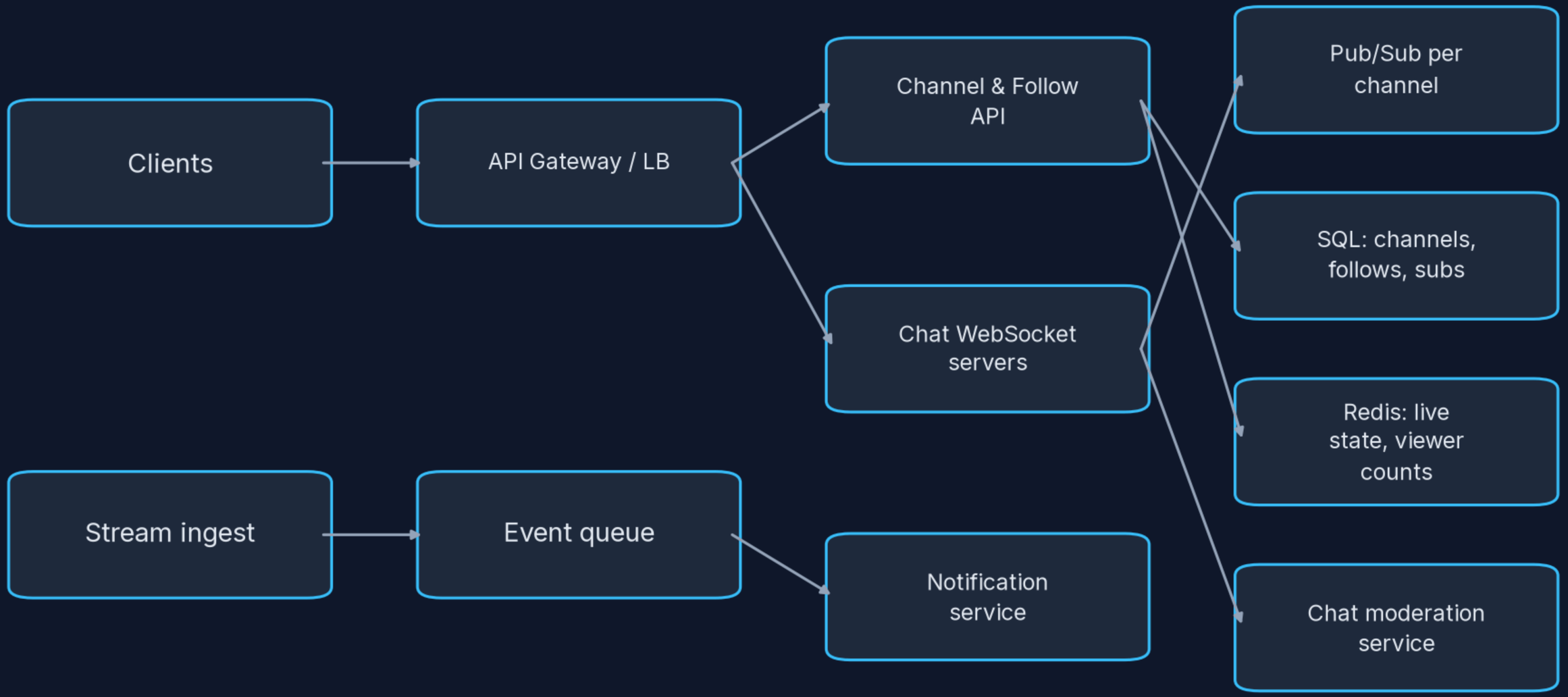
API

- Pagination with cursors on list endpoints.
- Auth with OAuth bearer tokens, scoped per action.

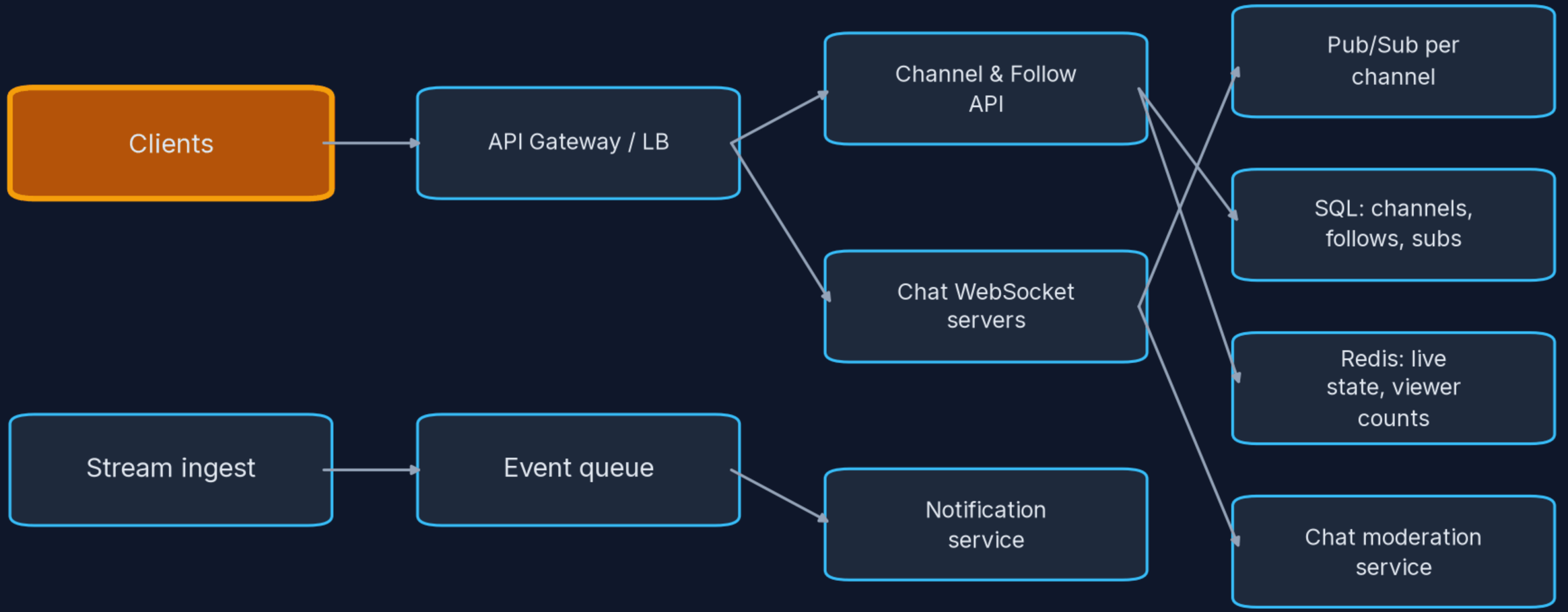
Capacity: worked out

- Big channel: 100k viewers, 50 msgs/s
- Fan-out: 50 × 100k = 5M deliveries/s
- Server holds ~50k sockets → ≥2 servers, each 2.5M pushes/s
- Batch every 100 ms → 10 frames/s per viewer

High-level architecture

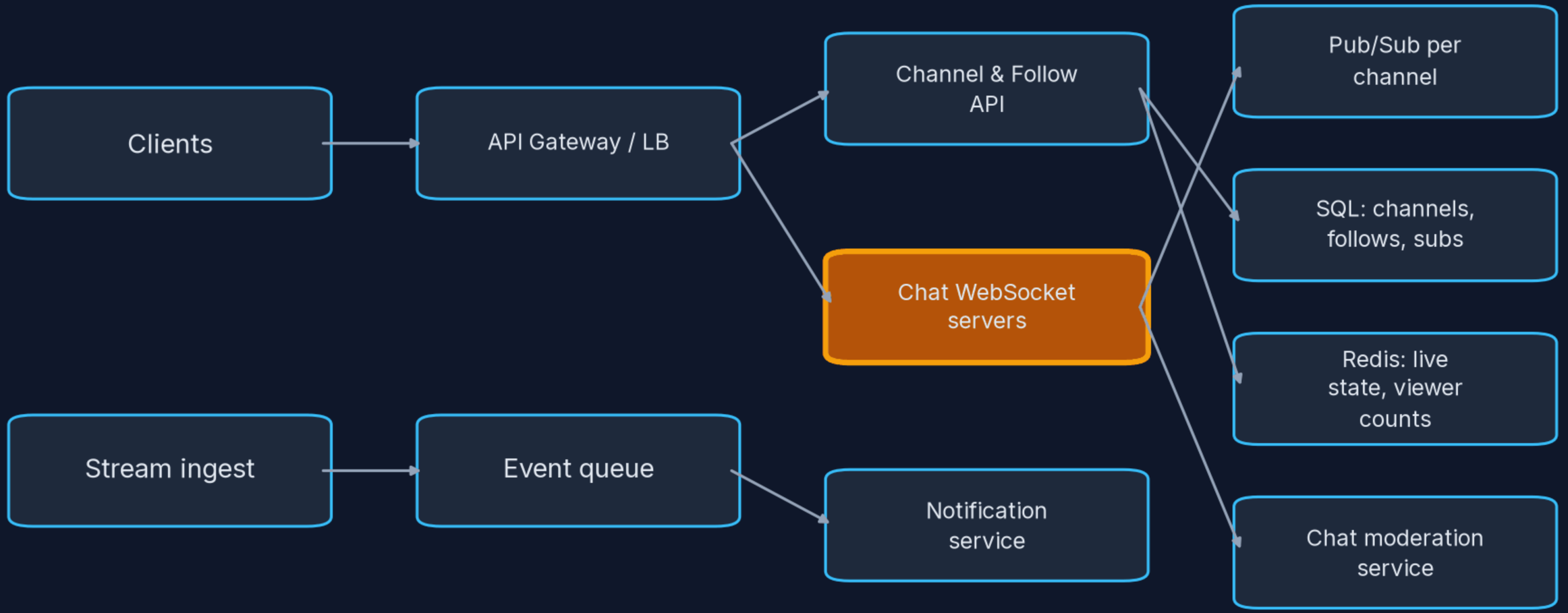


Request flow: Send a chat message



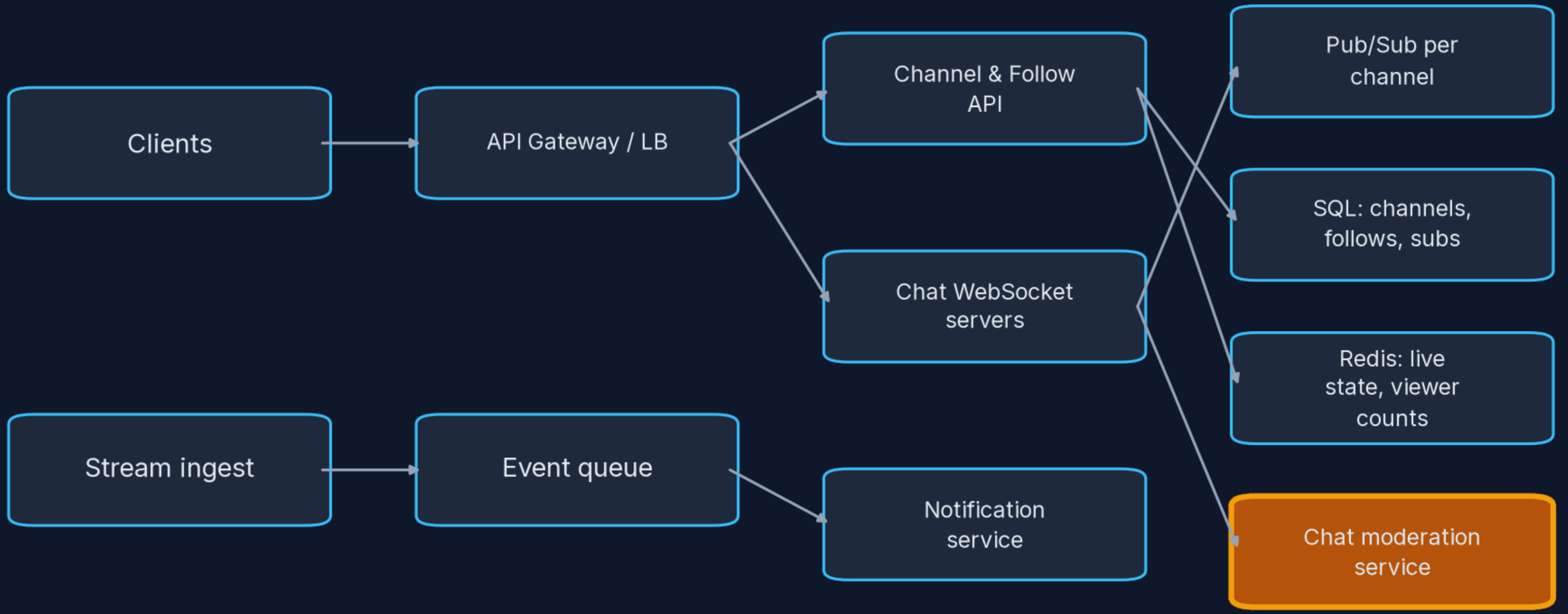
1. Viewer sends a chat message

Request flow: Send a chat message



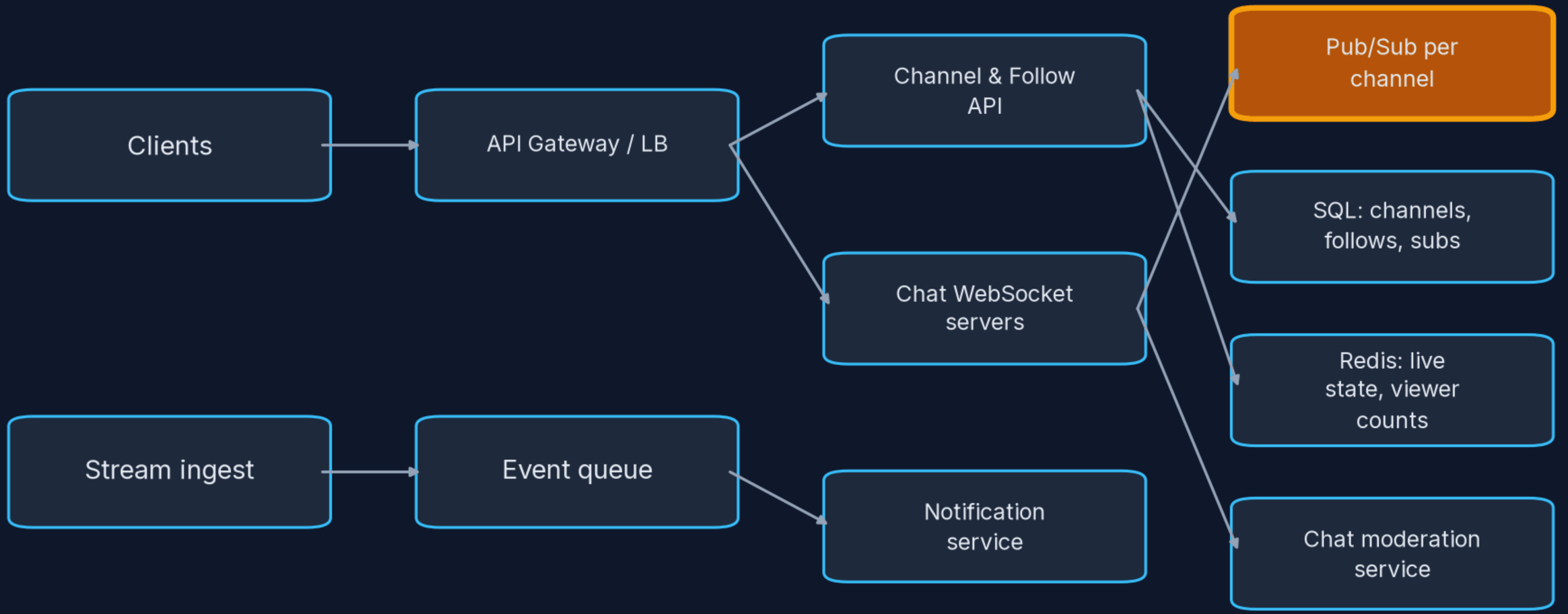
2. Chat server receives it

Request flow: Send a chat message



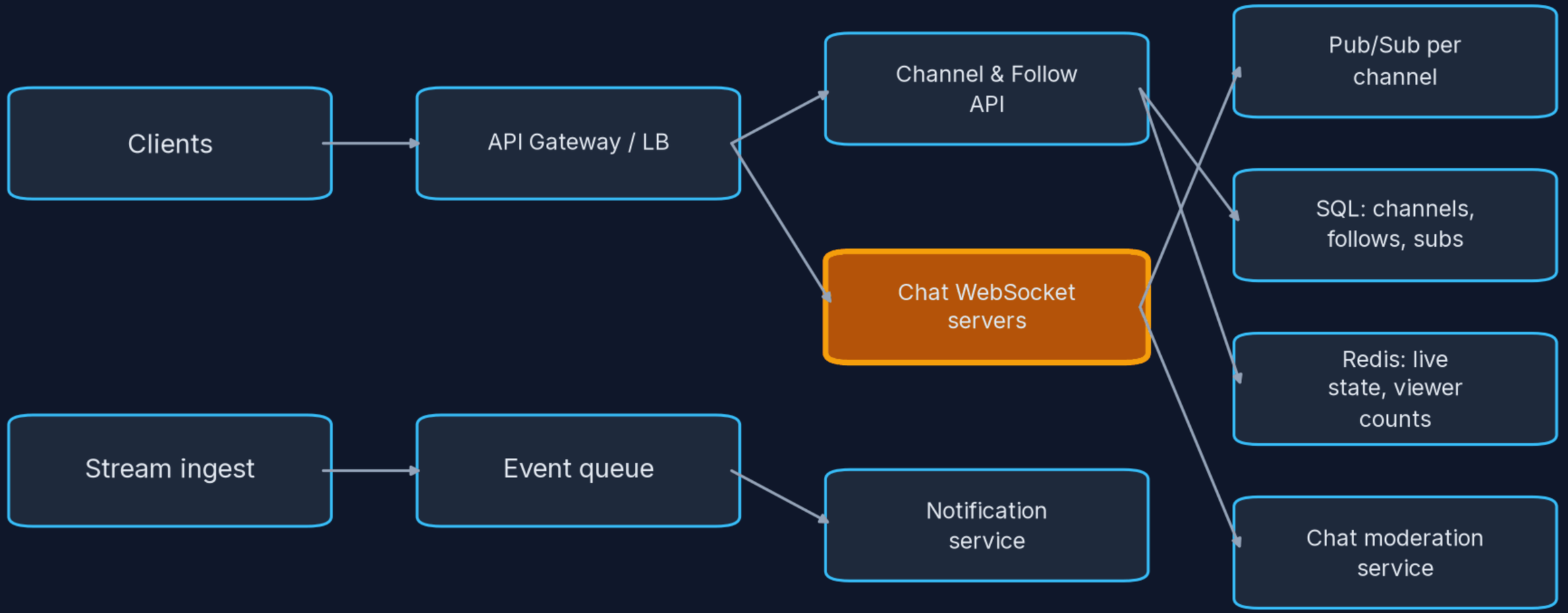
3. Moderation check

Request flow: Send a chat message



4. Publish to the channel topic

Request flow: Send a chat message



5. Fan out to viewers

Deep dives

- Chat fan-out: clients hash to WebSocket servers; each server subscribes to the channel topic it has viewers for. A message is published once and each server pushes to its local sockets. For huge channels, batch messages every ~100 ms and sample or rate-limit sending.
- Moderation: check bans and slow mode before publishing (cache them in Redis). Deletions are published as moderation events so clients remove messages.

Deep dives (cont.)

- Viewer count: each WS/edge server reports per-channel counts every few seconds; an aggregator sums them into Redis. Show approximate values.
- Go-live notifications: an ingest event goes onto a queue; workers page through followers and send push/email in batches. Large channels take minutes, which is acceptable.

Deep dives (cont.)

- Subscriptions: require an idempotency key, call the payment provider, write the subscription in a transaction, then emit an event (badge in chat, emote unlock).
- Video: the API returns only a signed CDN playlist URL (HLS); the video bytes never go through this API.

Key decisions

- Pub/sub per channel → publish once, fan out per server
- Approximate viewer counts → cheap, a few seconds stale is fine
- Strong consistency for subscriptions → money involved
- API serves playback URLs only → video bytes go via CDN

Trade-offs

- WebSockets versus long polling for chat: WS gives lower latency and less overhead.
- Chat is persisted asynchronously (log/queue → storage). The live path doesn't wait on the database.
- Viewer counts are eventually consistent; payments are strongly consistent.

Recap

- Clarify scope, then functional + non-functional needs
- Estimate load to size storage and fan-out
- Core pieces: Clients, API Gateway / LB, Channel & Follow API, Chat WebSocket servers, Pub/Sub per channel
- Know your trade-offs and failure modes